

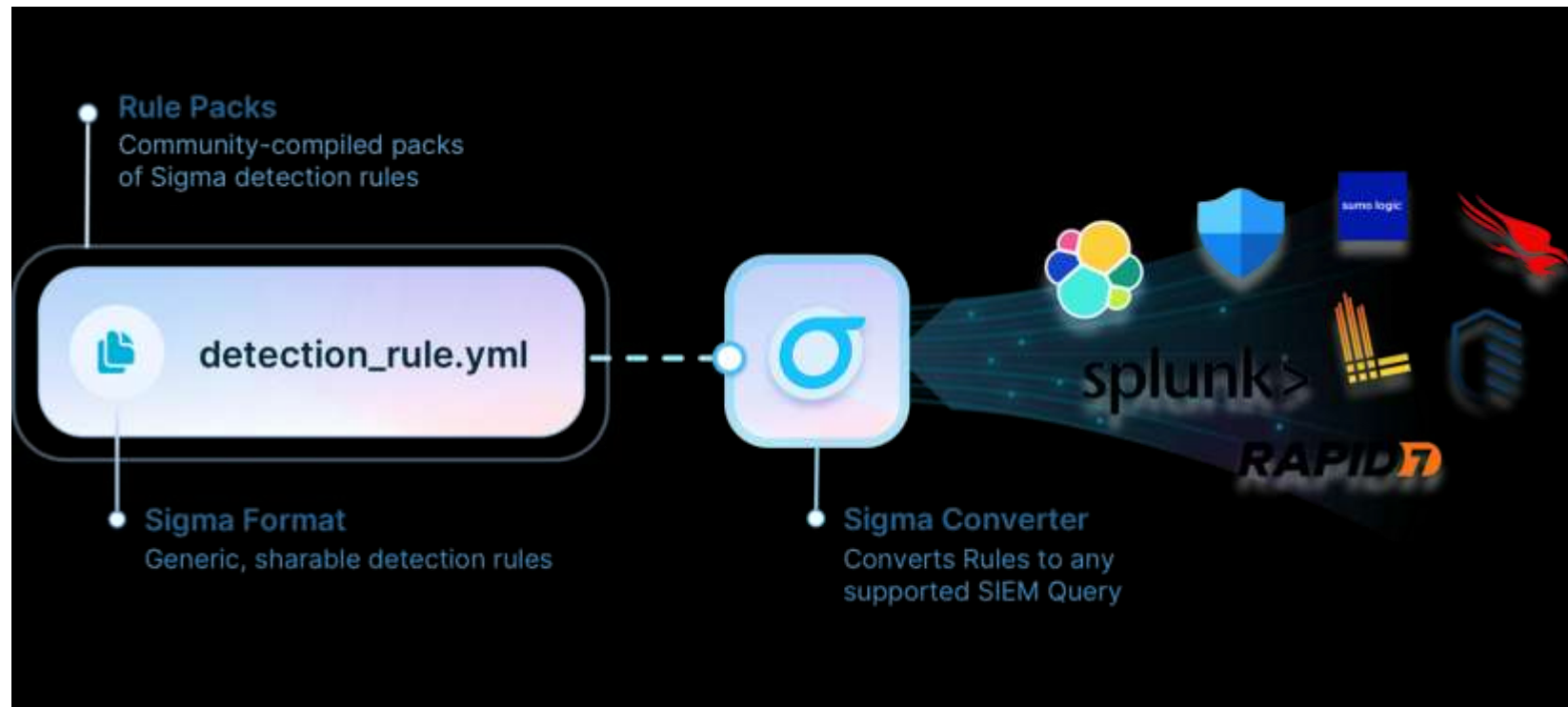
Live and Let Sigma Handout

Sigma Rules Workshop

1. Introduction to Sigma

*“Sigma is for log files
what Snort is for network traffic
and YARA is for files.”*

- Quote from sigmahq.io



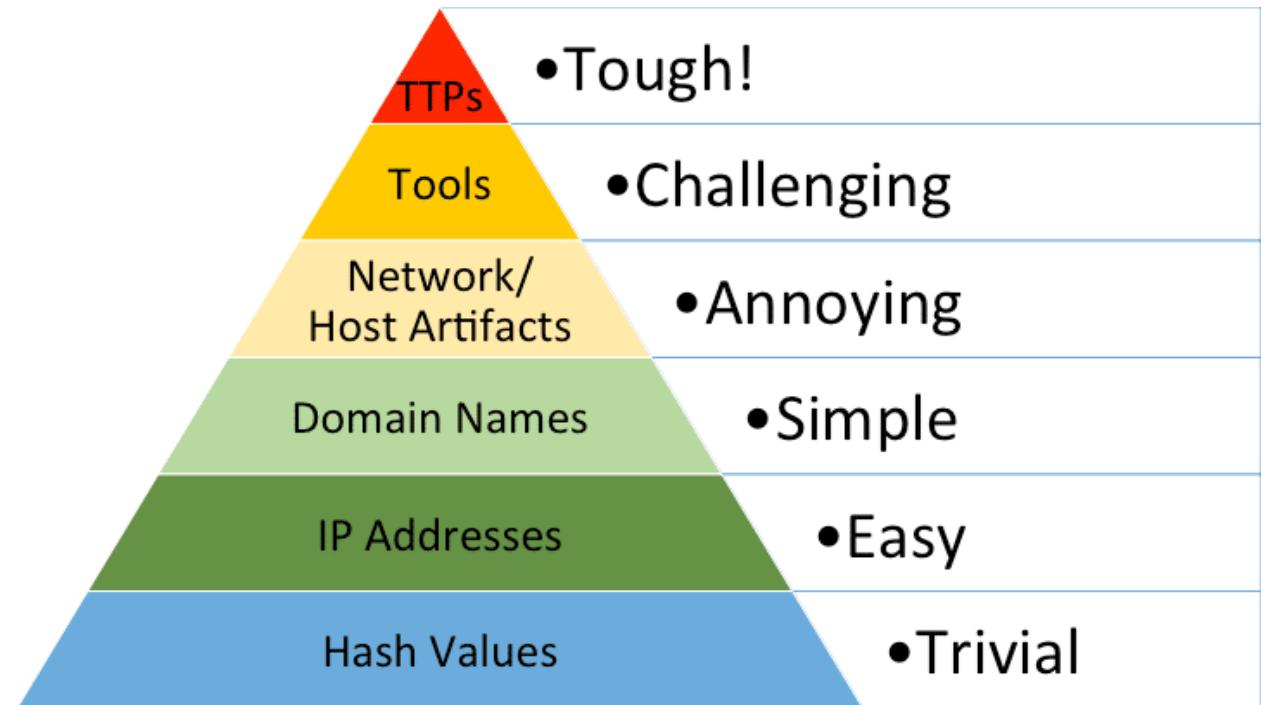
3. Sigma Converter – Quick demo

1. pip install sigma-cli (ALREADY done in your VM)
2. Open cmd.exe and go to `C:\custom_sigma_rules`
3. Test sigma-cli by using command: `sigma version` (output: 2.0.X)
4. Check your directory has sample sigma rule with name `interactive_at_job.yml`
5. `sigma convert --target splunk --pipeline splunk_windows
C:\custom_sigma_rules\interactive_at_job.yml`
6. You should output something like:

```
Image="*\at.exe" CommandLine="*interactive"
```

4. Detection Engineering Workflow:

1. Requirements Gathering and Threat Research
2. Rule Creation
3. Validation and Conversion
4. Testing and Tuning
5. Deployment and Monitoring



4.1 Detection Engineering Workflow:

1. Requirements Gathering and Threat Research
 1. Intel Analysis : Review Threat Reports, malware analysis, or incident data to identify specific behaviour or artifacts
 2. Tactics Mapping : Map activity to framework like MITRE ATT&CK
 3. Log Source Identification : Determine which logs capture necessary data
2. Rule Creation
3. Validation and Conversion
4. Testing and Tuning
5. Deployment and Monitoring

4.2 Detection Engineering Workflow:

1. Requirements Gathering and Threat Research
2. Rule Creation
 1. Metadata
 2. Logsource
 3. Detection Logic
3. Validation and Conversion
4. Testing and Tuning
5. Deployment and Monitoring

4.3 Detection Engineering Workflow:

1. Requirements Gathering and Threat Research
2. Rule Creation
3. Validation and Conversion
 1. Syntax Check
 2. Rule conversion
 3. Pipeline Application
4. Testing and Tuning
5. Deployment and Monitoring

4.4 Detection Engineering Workflow:

1. Requirements Gathering and Threat Research
2. Rule Creation
3. Validation and Conversion
4. Testing and Tuning
 1. Simulated Testing : Run converted query in historical or lab environment
 2. Performance Analysis: Check if query is too resource-intensive for SIEM
 3. Tuning : Adjust logic to reduce false positives
5. Deployment and Monitoring

4.5 Detection Engineering Workflow:

1. Requirements Gathering and Threat Research
2. Rule Creation
3. Validation and Conversion
4. Testing and Tuning
5. Deployment and Monitoring
 1. Version Control : Git Repository
 2. Lifecycle Management : Regular reviews to update or deprecate

Aurora EDR

1. What is Aurora ?
2. Lightweight agent that applies Sigma rules on log data in real-time on endpoints
3. Uses ETW (Event Tracing for Windows)
4. Managed locally via config files or via ASGARD Management Center
5. Consider it as custom Sigma-based EDR
6. Extends the Sigma standard with 'response' actions
 1. Kill, KillParent, Suspend, Dump
 2. Custom actions

Aurora EDR: **specific Response Actions**

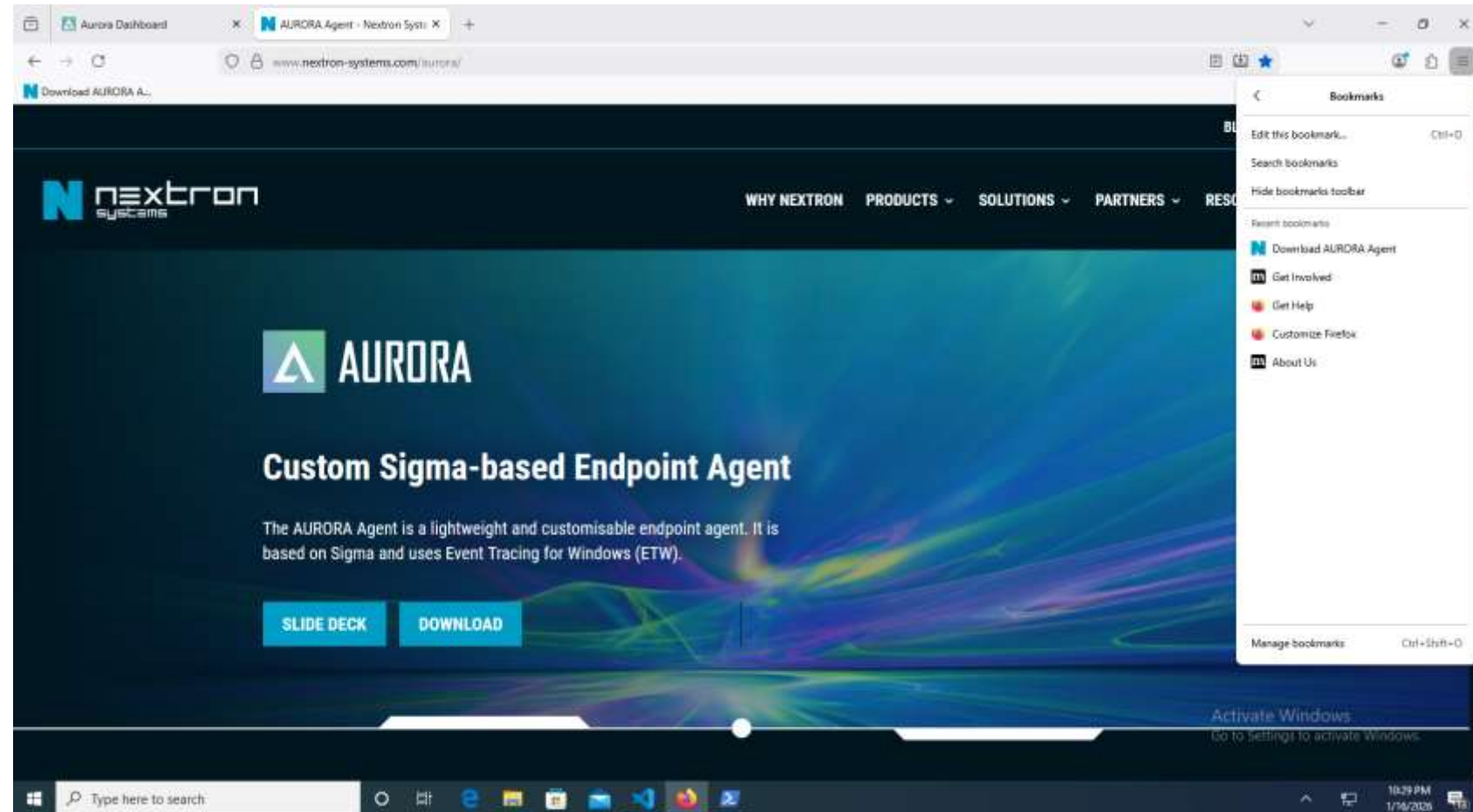
1. In the Sigma rule we can specify response action for Aurora EDR
2. Response action can be:
 1. Predefined
 1. Kill a process or parent process
 2. Suspend a process
 3. Dump Process memory
 2. Custom
 1. A custom command line that can use environment variables and event values
 2. e.g. `copy %image% %%ProgramData%%\ProcessId%.bin`
3. This can help containment of threat quickly

Installation of Aurora Agent

1. In your Windows 10 VM open C:\ Folder
2. You should see a folder **aurora-agent-lite-win-pack**
3. **Only If you don't the folder follow rest of steps 4,5**
4. You should find zip file with name **aurora-agent-lite-win-pack.zip**
 - i. If you do not find above file do get the USB drive and copy this file in your VM
5. Unzip the **aurora-agent-lite-win-pack.zip** file by right click and selecting Extract All menu.
6. Check if c:\ have 4 more folders **attack**, **custom_sigma_rules**, **sigma_test_repo**, **web**
7. In Explorer go inside the folder **C:\aurora-agent-lite-win-pack**

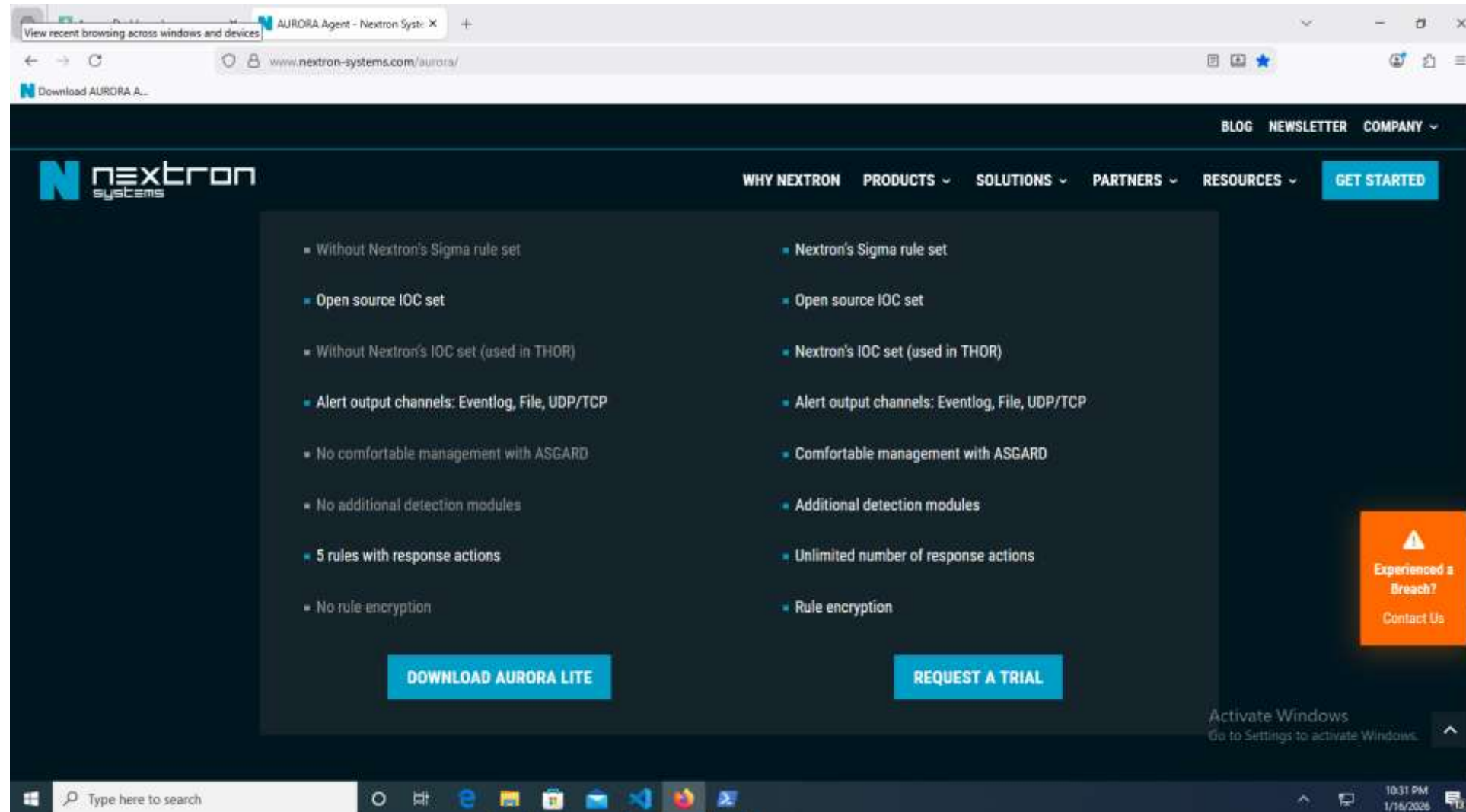
Installation of Aurora Agent

- Open following link: <https://www.nextron-systems.com/aurora/>
- There should be bookmark “Download AURORA Agent” in your browser for this not open by typing URL
- Click Download



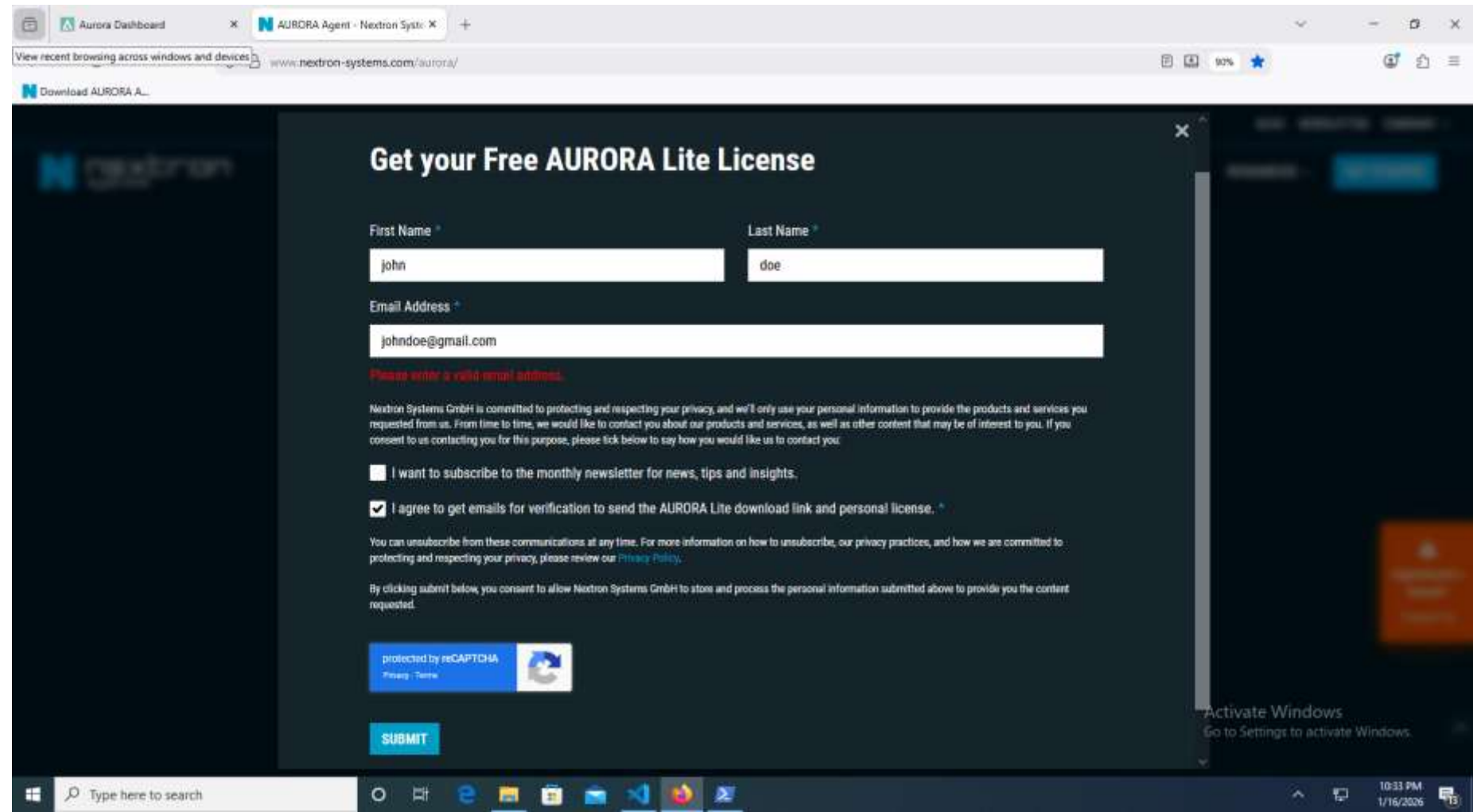
Installation of Aurora Agent

- Click on button DOWNLOAD AURORA LITE



Installation of Aurora Agent

- Fill in any name but use email where you can get email
- You can use disposable email also if you want



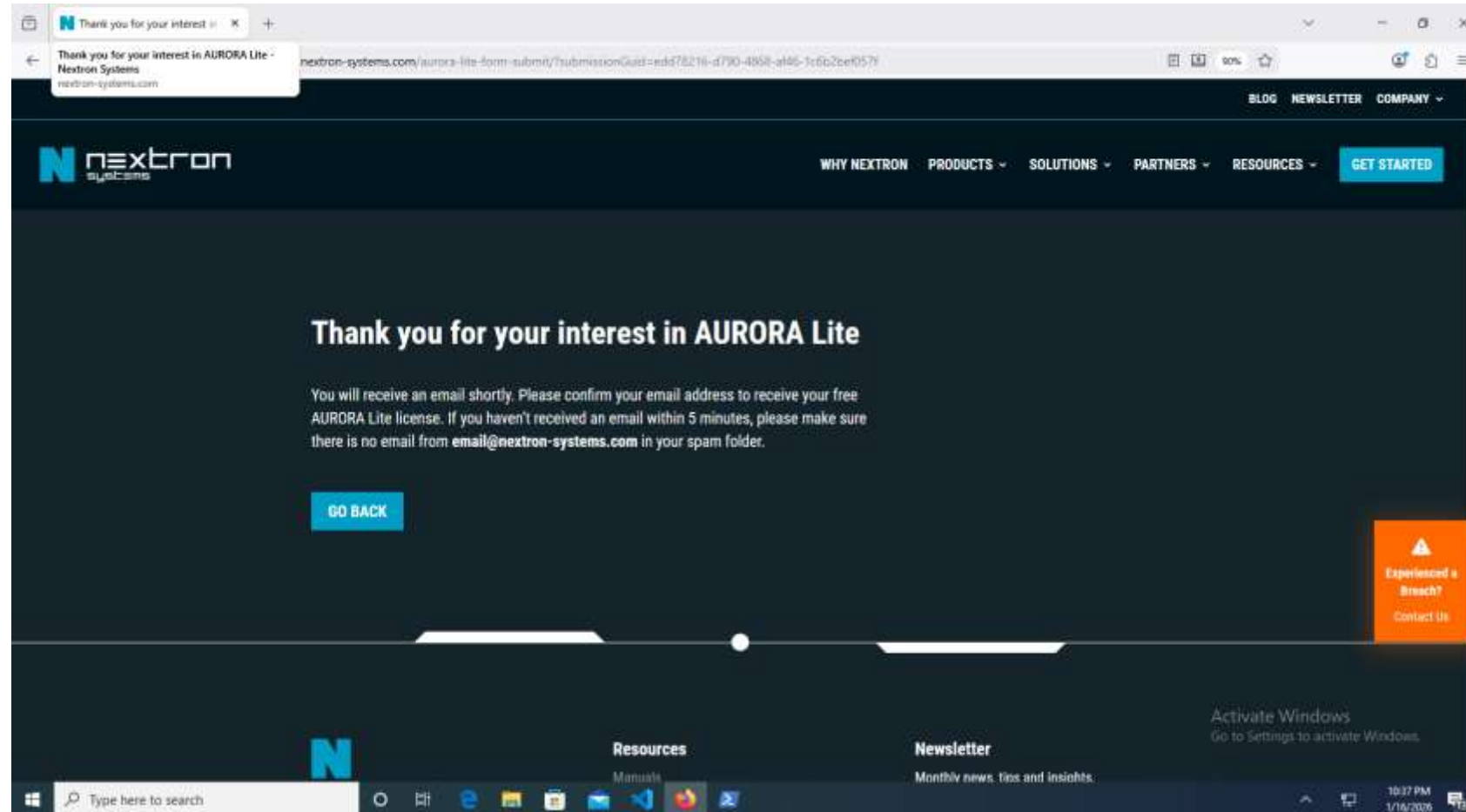
The screenshot shows a web browser window with the URL www.nexttron-systems.com/aurora/. The page is titled "Get your Free AURORA Lite License". It contains a registration form with the following fields and content:

- First Name ***: Input field containing "john".
- Last Name ***: Input field containing "doe".
- Email Address ***: Input field containing "johndoe@gmail.com". Below this field is a red error message: "Please enter a valid email address."
- Consent Section**:
 - A paragraph: "Nexttron Systems GmbH is committed to protecting and respecting your privacy, and we'll only use your personal information to provide the products and services you requested from us. From time to time, we would like to contact you about our products and services, as well as other content that may be of interest to you. If you consent to us contacting you for this purpose, please tick below to say how you would like us to contact you."
 - Two checkboxes:
 - ☐ I want to subscribe to the monthly newsletter for news, tips and insights.
 - ☒ I agree to get emails for verification to send the AURORA Lite download link and personal license. *
 - A paragraph: "You can unsubscribe from these communications at any time. For more information on how to unsubscribe, our privacy practices, and how we are committed to protecting and respecting your privacy, please review our [Privacy Policy](#)."
 - A paragraph: "By clicking submit below, you consent to allow Nexttron Systems GmbH to store and process the personal information submitted above to provide you the content requested."
- Submit Button**: A blue button labeled "SUBMIT".
- reCAPTCHA**: A blue box indicating "protected by reCAPTCHA" with links for "Privacy" and "Terms".

The Windows taskbar at the bottom shows the time as 10:33 PM on 1/16/2025.

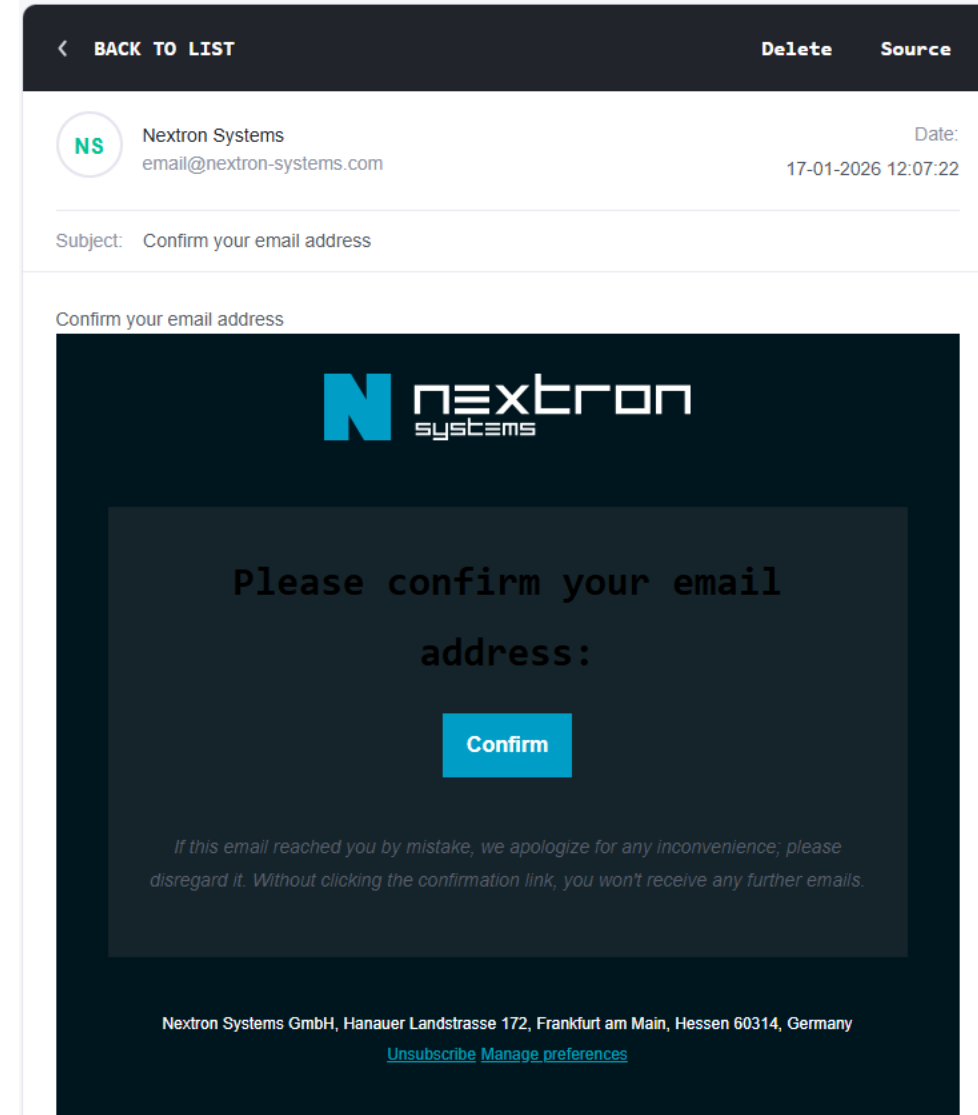
Installation of Aurora Agent

- You Should see something like this Thank You message:



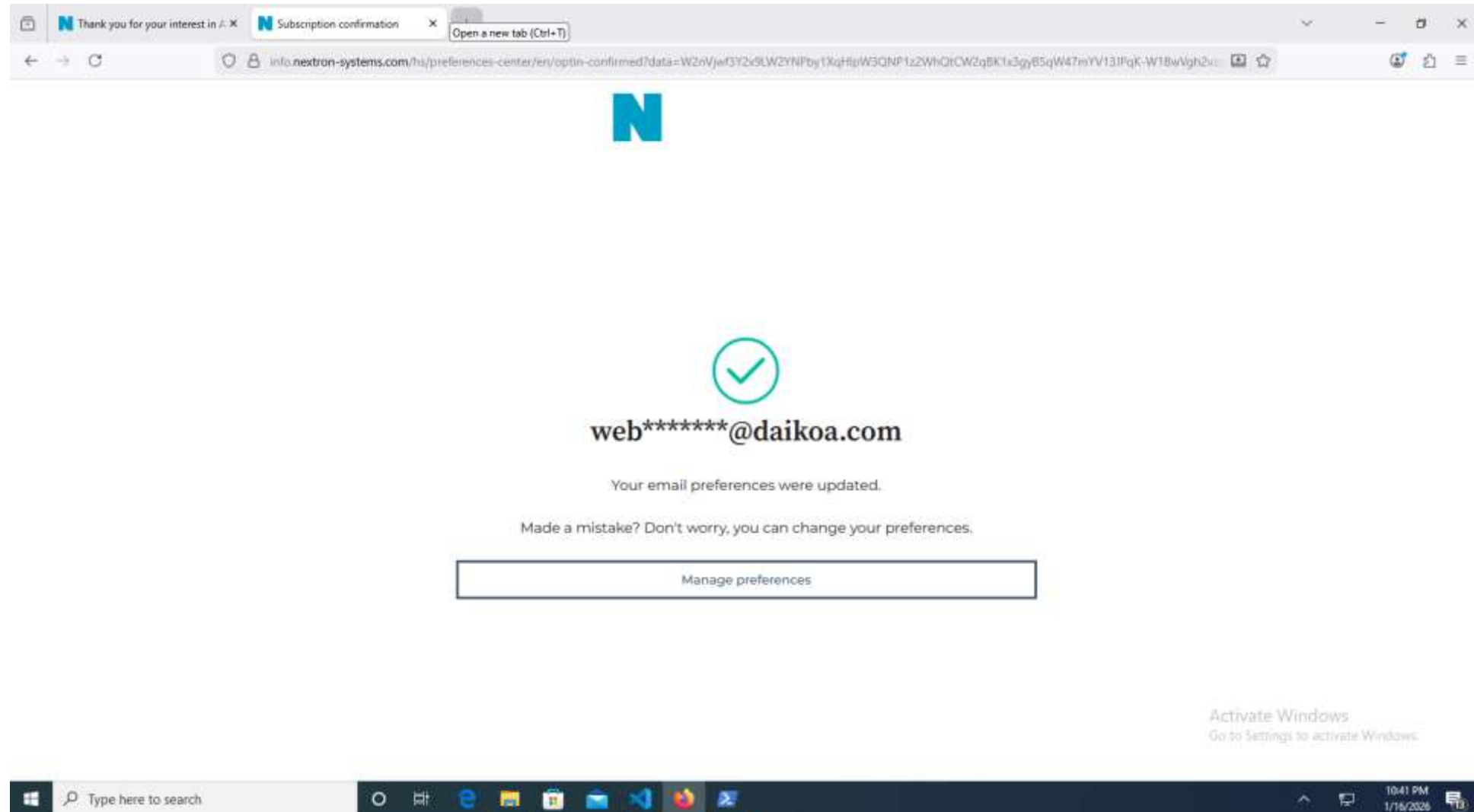
Installation of Aurora Agent

- You will receive a email like this click on “Confirm” button



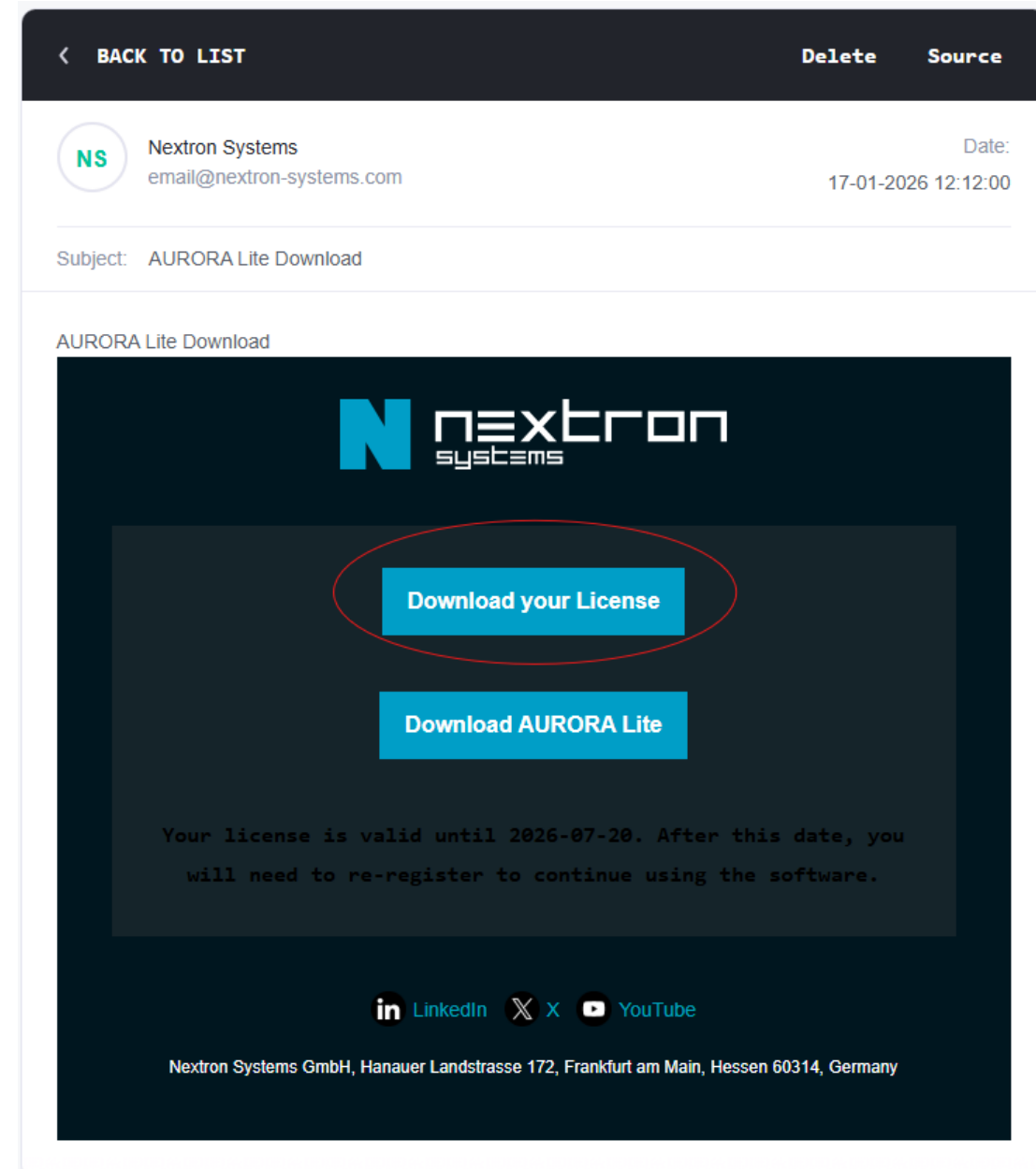
Installation of Aurora Agent

- You will receive a email like this click on “Confirm” button



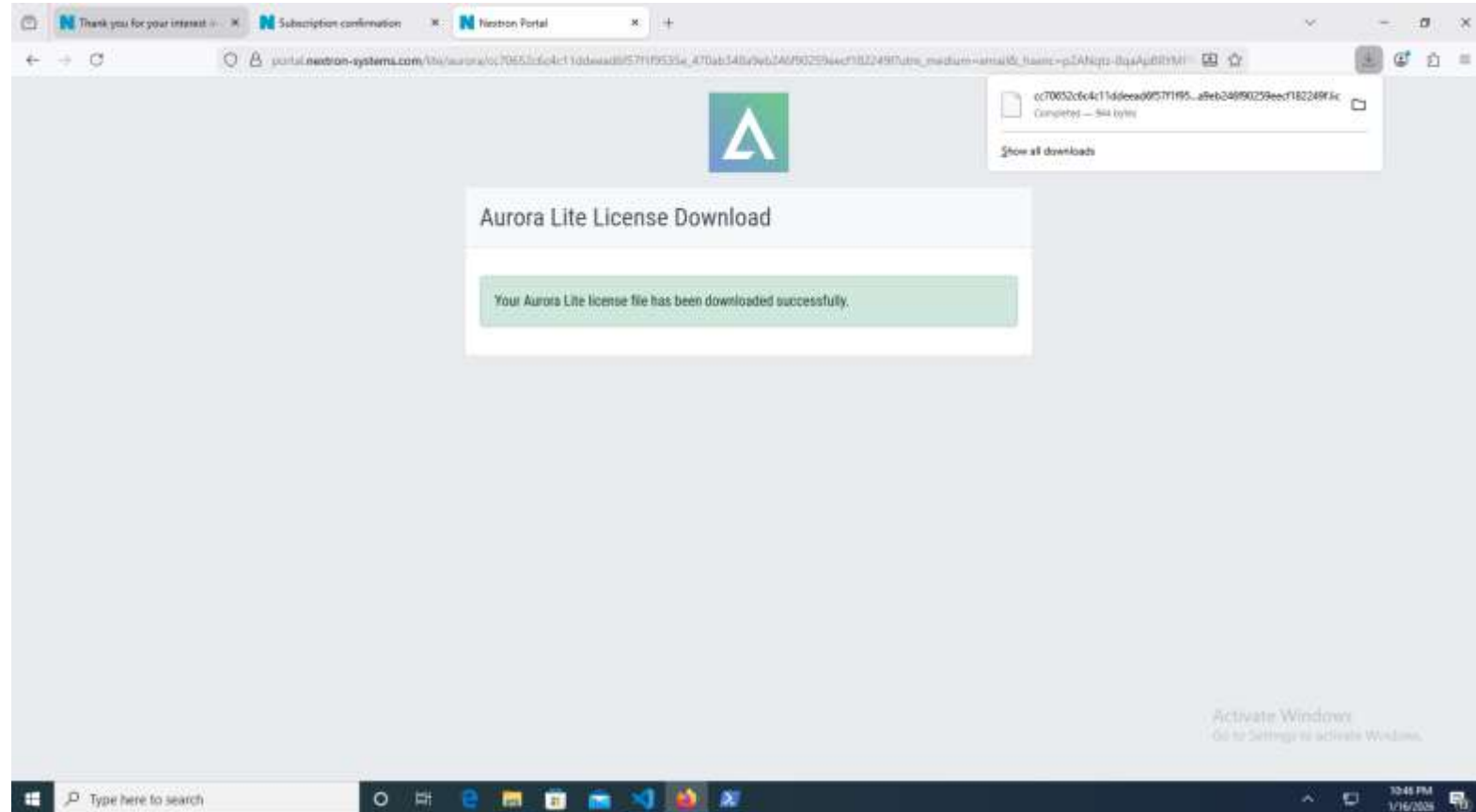
Installation of Aurora Agent

- You will receive a email with subject: AURORA Lite Download
- Click on Download your Licence
- If you are outside your VM (on host) for internet access then copy this .lic file to your Windows 10 VM



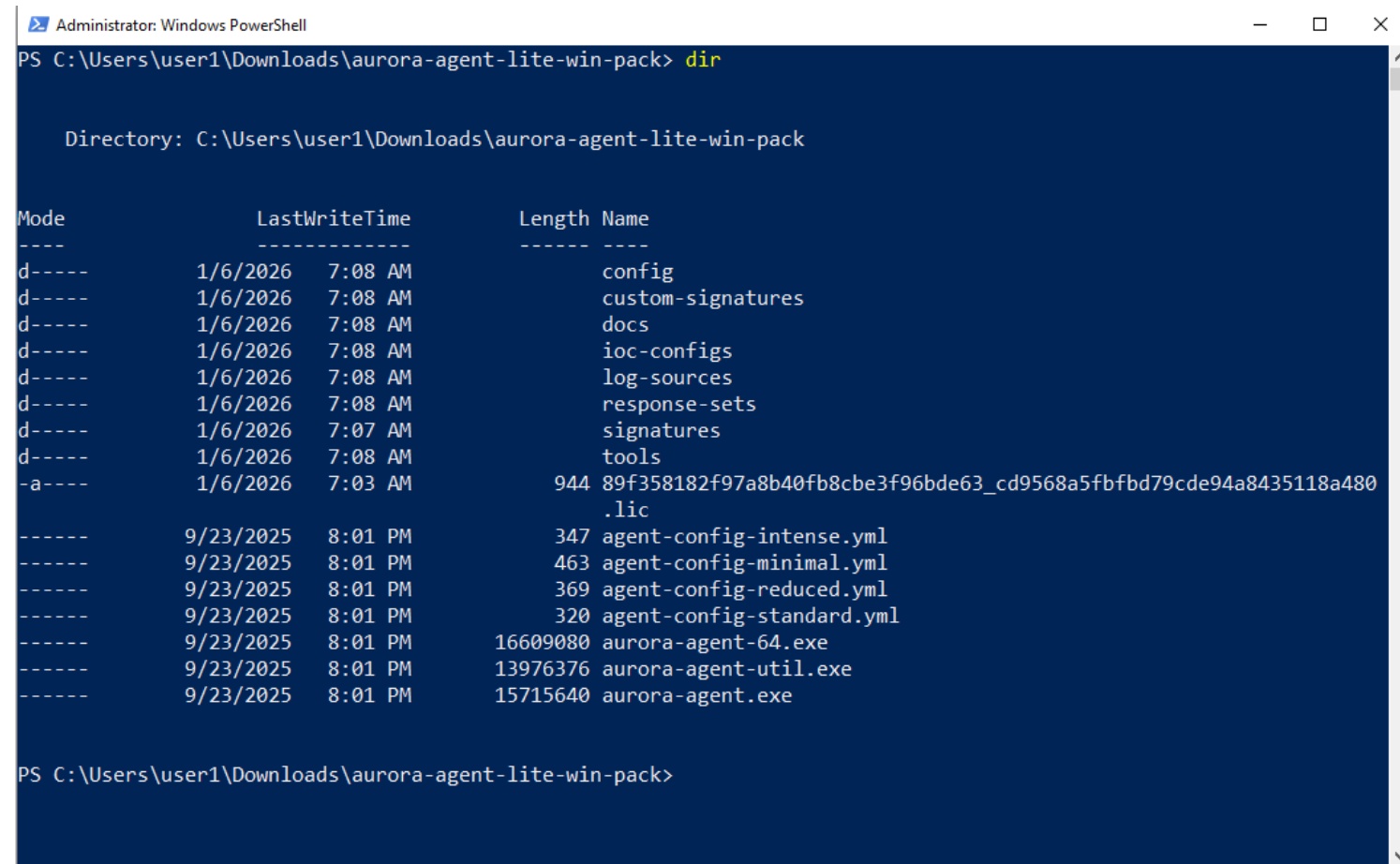
Installation of Aurora Agent

- Copy the downloaded .lic file in to the extracted AURORA folder in Downloads directory
- C:\aurora-agent-lite-win-pack\



Installation of Aurora Agent

- In Windows Explorer Click on File and Open Window PowerShell -> Open Window PowerShell as Administrator
- Using “dir” command verify .lic file is in same extracted folder



```
Administrator: Windows PowerShell
PS C:\Users\user1\Downloads\aurora-agent-lite-win-pack> dir

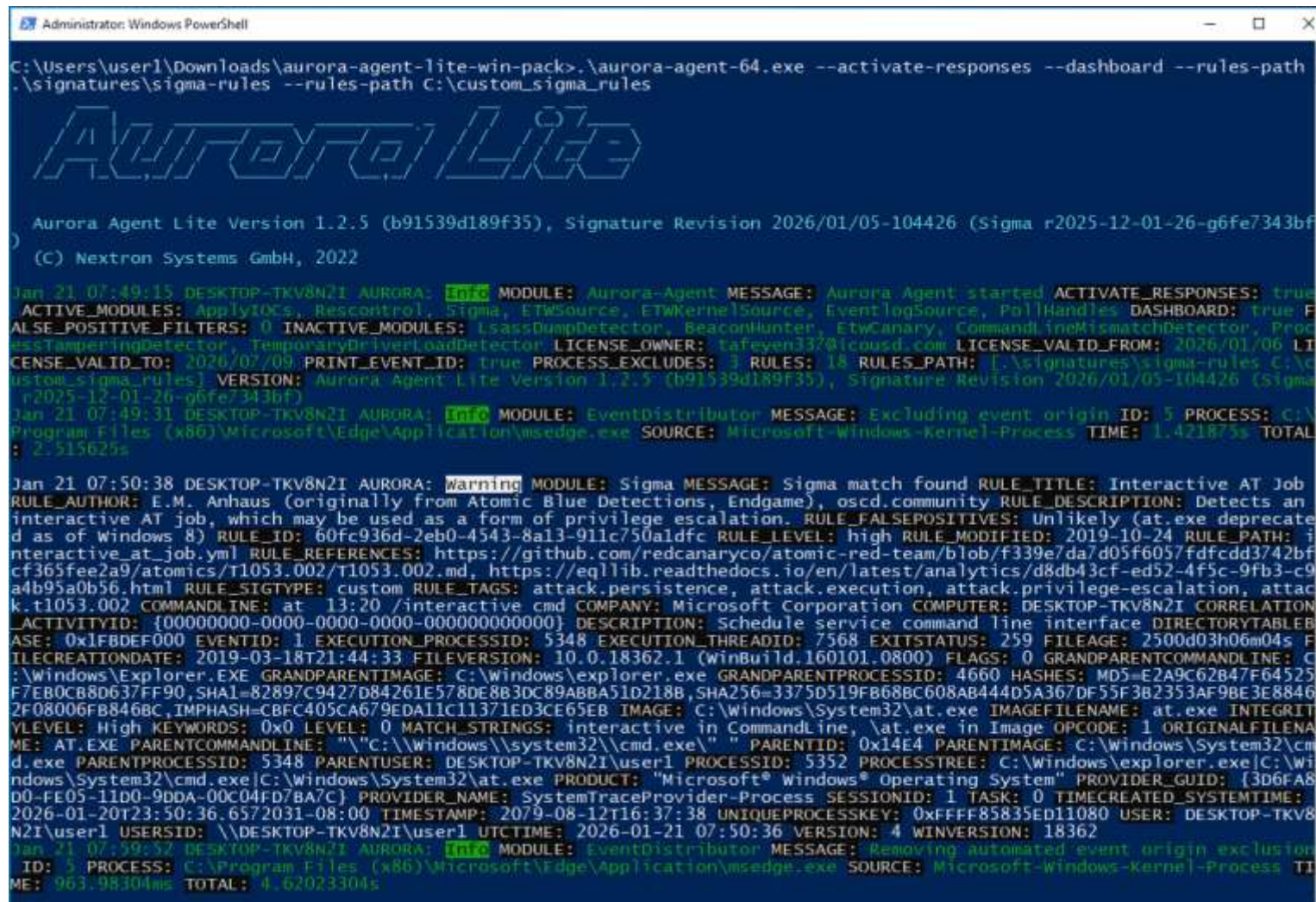
Directory: C:\Users\user1\Downloads\aurora-agent-lite-win-pack

Mode                LastWriteTime         Length Name
----                -
d-----          1/6/2026   7:08 AM                config
d-----          1/6/2026   7:08 AM            custom-signatures
d-----          1/6/2026   7:08 AM                docs
d-----          1/6/2026   7:08 AM            ioc-configs
d-----          1/6/2026   7:08 AM            log-sources
d-----          1/6/2026   7:08 AM            response-sets
d-----          1/6/2026   7:07 AM            signatures
d-----          1/6/2026   7:08 AM                tools
-a----          1/6/2026   7:03 AM           944 89f358182f97a8b40fb8cbe3f96bde63_cd9568a5fbfd79cde94a8435118a480
               .lic
-----          9/23/2025   8:01 PM           347 agent-config-intense.yml
-----          9/23/2025   8:01 PM           463 agent-config-minimal.yml
-----          9/23/2025   8:01 PM           369 agent-config-reduced.yml
-----          9/23/2025   8:01 PM           320 agent-config-standard.yml
-----          9/23/2025   8:01 PM       16609080 aurora-agent-64.exe
-----          9/23/2025   8:01 PM       13976376 aurora-agent-util.exe
-----          9/23/2025   8:01 PM       15715640 aurora-agent.exe

PS C:\Users\user1\Downloads\aurora-agent-lite-win-pack>
```

Installation of Aurora Agent

- Run Agent by using command:
 `.\start_custom_agent.bat`
- You should see output like:



```
Administrator: Windows PowerShell
C:\Users\user1\Downloads\aurora-agent-lite-win-pack>.\aurora-agent-64.exe --activate-responses --dashboard --rules-path
.\signatures\sigma-rules --rules-path C:\custom_sigma_rules

Aurora Lite

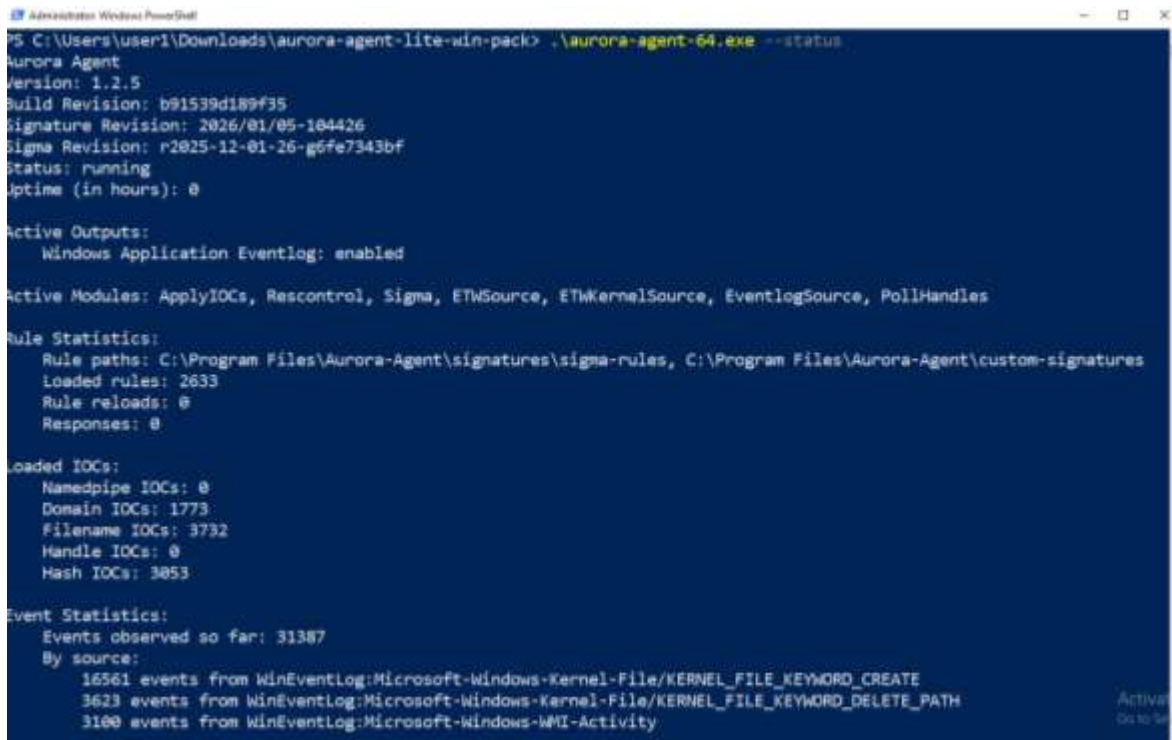
Aurora Agent Lite Version 1.2.5 (b91539d189f35), Signature Revision 2026/01/05-104426 (Sigma r2025-12-01-26-g6fe7343bf)
(c) Nextron Systems GmbH, 2022

Jan 21 07:49:15 DESKTOP-TKV8N2I AURORA: [Info] MODULE: Aurora-Agent MESSAGE: Aurora Agent started/ACTIVATE_RESPONSES: true
ACTIVE_MODULES: ApplyIOCs, Rescontrol, Sigma, ETWSource, ETWKernelSource, EventlogSource, PollHandles DASHBOARD: true F
ALSE_POSITIVE_FILTERS: 0 INACTIVE_MODULES: LsassDumpDetector, BeaconHunter, EtwCanary, CommandlineMismatchDetector, Proc
essTamperingDetector, TemporaryDriverLoadDetector LICENSE_OWNER: tafeyen332@icloud.com LICENSE_VALID_FROM: 2026/01/06 LI
CENSE_VALID_TO: 2026/07/09 PRINT_EVENT_ID: true PROCESS_EXCLUDES: 3 RULES: 18 RULES_PATH: [.\signatures\sigma-rules C:\c
ustom_sigma_rules] VERSION: Aurora Agent Lite Version 1.2.5 (b91539d189f35), Signature Revision 2026/01/05-104426 (Sigma
r2025-12-01-26-g6fe7343bf)
Jan 21 07:49:31 DESKTOP-TKV8N2I AURORA: [Info] MODULE: EventDistributor MESSAGE: Excluding event origin ID: 5 PROCESS: C:\
Program Files (x86)\Microsoft\Edge\Application\msedge.exe SOURCE: Microsoft-Windows-Kernel-Process TIME: 1.421875s TOTAL
: 2.515625s

Jan 21 07:50:38 DESKTOP-TKV8N2I AURORA: [Warning] MODULE: Sigma MESSAGE: Sigma match found RULE_TITLE: Interactive AT Job
RULE_AUTHOR: E.M. Anhaus (originally from Atomic Blue Detections, Endgame), oscd.community RULE_DESCRIPTION: Detects an
interactive AT job, which may be used as a form of privilege escalation. RULE_FALSEPOSITIVES: Unlikely (at.exe deprecate
d as of windows 8) RULE_ID: 60fc936d-2eb0-4543-8a13-911c750a1dfc RULE_LEVEL: high RULE_MODIFIED: 2019-10-24 RULE_PATH: i
nteractive_at.job.yml RULE_REFERENCES: https://github.com/redcanaryco/atomic-red-team/blob/f339e7da7d05f6057fdcfdd3742bf
cf365fee2a9/atomics/T1053.002/T1053.002.md, https://eqllib.readthedocs.io/en/latest/analytics/d8db43cf-ed52-4f5c-9fb3-c9
a4b95a0b56.html RULE_SIGTYPE: custom RULE_TAGS: attack.persistence, attack.execution, attack.privilege-escalation, attac
k.t1053.002 COMMANDLINE: at 13:20 /interactive cmd COMPANY: Microsoft Corporation COMPUTER: DESKTOP-TKV8N2I CORRELATION
_ACTIVITYID: {00000000-0000-0000-0000-000000000000} DESCRIPTION: Schedule service command line interface DIRECTORYTABLER
ASE: 0x1FBDEF000 EVENTID: 1 EXECUTION_PROCESSID: 5348 EXECUTION_THREADID: 7568 EXITSTATUS: 259 FILEAGE: 2500d03h06m04s F
ILECREATIONDATE: 2019-03-18T21:44:33 FILEVERSION: 10.0.18362.1 (winBuild.160101.0800) FLAGS: 0 GRANDPARENTCOMMANDLINE: C
:\Windows\Explorer.EXE GRANDPARENTIMAGE: C:\Windows\explorer.exe GRANDPARENTPROCESSID: 4660 HASHES: MD5=E2A9C62B47F64525
F7EB0CB8D637FF90, SHA1=82897C9427D84261E578DE8B3DC89ABBA51D218B, SHA256=3375D519FB68BC608AB44D5A367DF55F3B2353AF9BE3E8846
2F08006FB846BC, IMPHASH=CBFC405CA679EDA11C11371ED3CE65EB IMAGE: C:\Windows\System32\at.exe IMAGEFILENAME: at.exe INTEGRIT
YLEVEL: High KEYWORDS: 0x0 LEVEL: 0 MATCH_STRINGS: interactive in CommandLine, \at.exe in Image OPCode: 1 ORIGINALFILENA
ME: AT.EXE PARENTCOMMANDLINE: "\"C:\Windows\system32\cmd.exe\" \" PARENTID: 0x14E4 PARENTIMAGE: C:\Windows\System32\cm
d.exe PARENTPROCESSID: 5348 PARENTUSER: DESKTOP-TKV8N2I\user1 PROCESSID: 5352 PROCSSTREE: C:\Windows\explorer.exe|C:\Wi
ndows\System32\cmd.exe|C:\Windows\System32\at.exe PRODUCT: "Microsoft® Windows® Operating System" PROVIDER_GUID: {3D6FA8
D0-FE05-11D0-9DDA-00C04FD7BA7C} PROVIDER_NAME: SystemTraceProvider-Process SESSIONID: 1 TASK: 0 TIMECREATED_SYSTEMTIME:
2026-01-20T23:50:36.6572031-08:00 TIMESTAMP: 2079-08-12T16:37:38 UNIQUEPROCESSKEY: 0xFFFF85835ED11080 USER: DESKTOP-TKV8
N2I\user1 USERSID: \\DESKTOP-TKV8N2I\user1 UTCTIME: 2026-01-21 07:50:36 VERSION: 4 WINVERSION: 18362
Jan 21 07:59:52 DESKTOP-TKV8N2I AURORA: [Info] MODULE: EventDistributor MESSAGE: Removing automated event origin exclusion
ID: 5 PROCESS: C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe SOURCE: Microsoft-Windows-Kernel-Process TI
ME: 963.98304ms TOTAL: 4.62023304s
```


Installation of Aurora Agent

- Open new PowerShell window and Verify installation by using command:
 `.\aurora-agent64.exe --status`
- Verify new events in the local “Application” event log (Event Viewer)
 - Win+R --type--> eventvwr.msc
 - EventViwer (Local) -> Windows Logs -> Application



```
PS C:\Users\user1\Downloads\aurora-agent-lite-win-pack> .\aurora-agent-64.exe --status
Aurora Agent
Version: 1.2.5
Build Revision: b91539d189f35
Signature Revision: 2026/01/05-104426
Sigma Revision: r2025-12-01-26-g6fe7343bf
Status: running
Uptime (in hours): 0

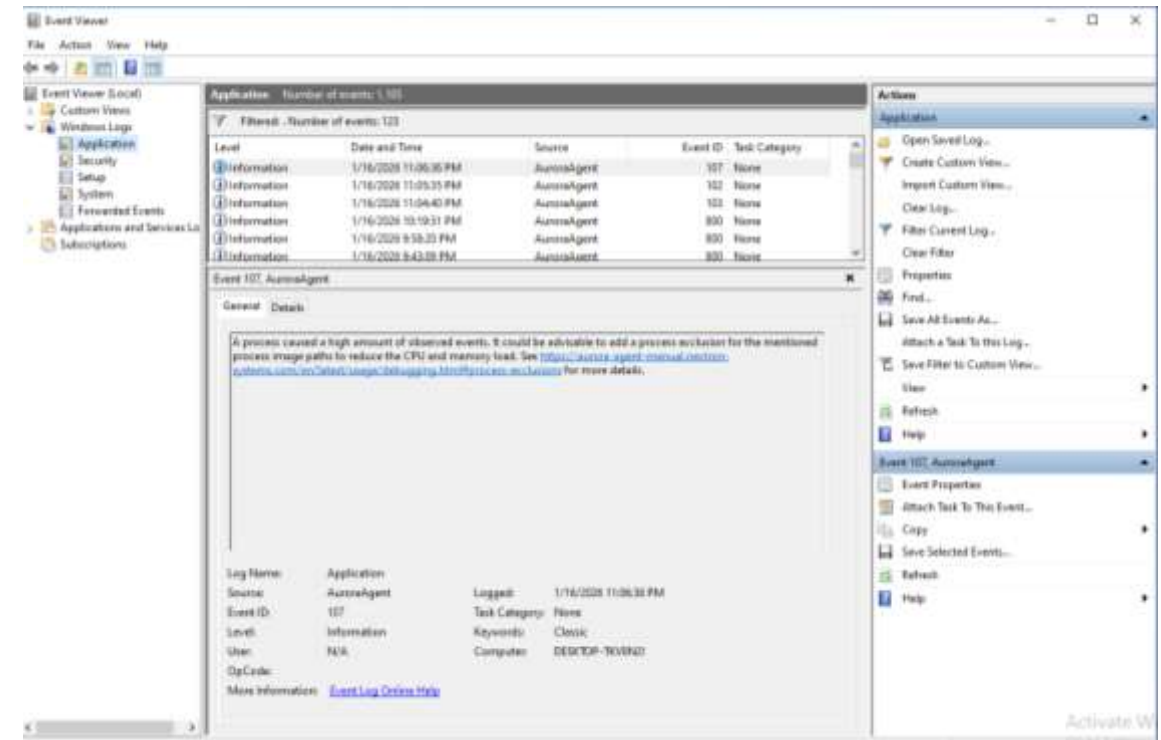
Active Outputs:
  Windows Application Eventlog: enabled

Active Modules: ApplyIOCs, Rescontrol, Sigma, ETHSource, ETHKernelSource, EventlogSource, PollHandles

Rule Statistics:
  Rule paths: C:\Program Files\Aurora-Agent\signatures\sigma-rules, C:\Program Files\Aurora-Agent\custom-signatures
  Loaded rules: 2633
  Rule reloads: 0
  Responses: 0

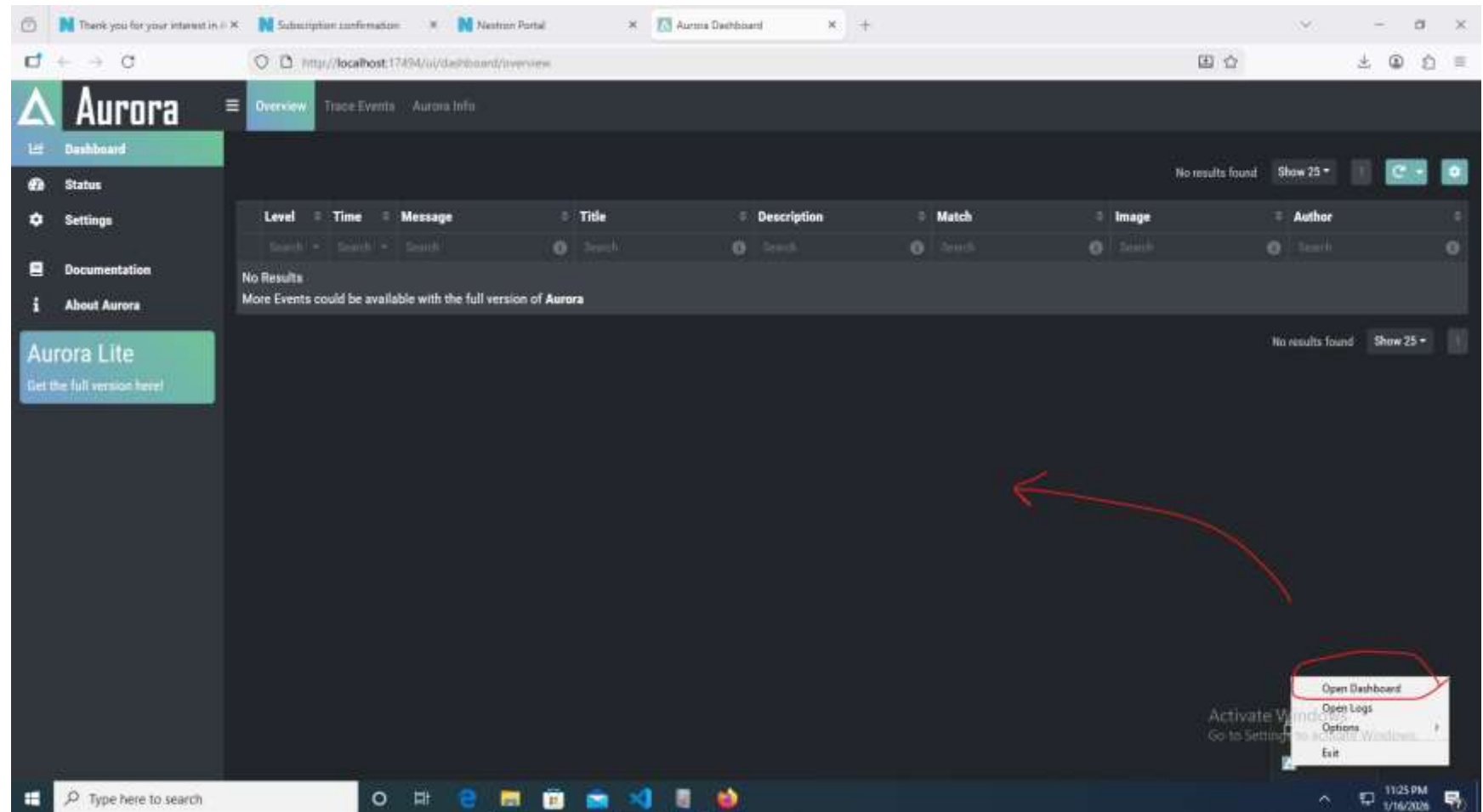
Loaded IOCs:
  Namedpipe IOCs: 0
  Domain IOCs: 1773
  Filename IOCs: 3732
  Handle IOCs: 0
  Hash IOCs: 3053

Event Statistics:
  Events observed so far: 31387
  By source:
    16561 events from WinEventLog:Microsoft-Windows-Kernel-File/KERNEL_FILE_KEYWORD_CREATE
    3623 events from WinEventLog:Microsoft-Windows-Kernel-File/KERNEL_FILE_KEYWORD_DELETE_PATH
    3100 events from WinEventLog:Microsoft-Windows-WMI-Activity
```



Installation of Aurora Agent

- Right click on aurora icon near bottom right corner
- Click on menu



Installation of Aurora Agent

- If default click on dashboard does not open the browser do following:
 - Open Edge/ Firefox browser
 - In browser open URL
`http://localhost:17494/ui/dashboard/overview`

Installation of Aurora Agent

Status Tab:

The screenshot shows the Aurora Agent Status Tab interface in a web browser. The browser's address bar displays `http://localhost:17434/ui/status/overview`. The interface features a dark theme with a sidebar on the left containing navigation links: Dashboard, Status (highlighted), Settings, Documentation, and About Aurora. Below these links is a green button labeled "Aurora Lite" with the text "Get the full version here!".

The main content area is divided into four panels:

- Aurora Agent Info:** A table displaying agent details.

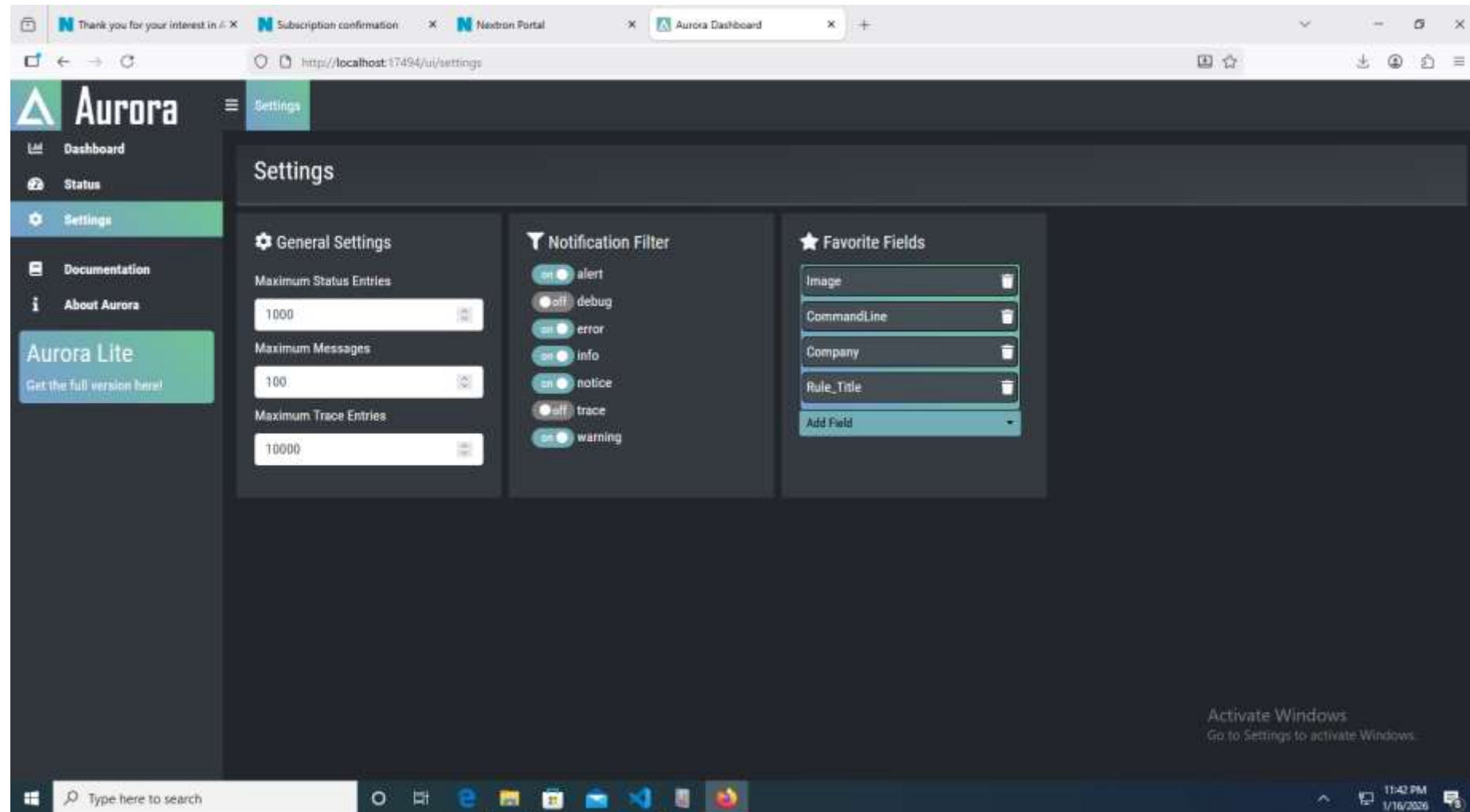
Property	Value
Product	Aurora Agent Lite
Version	1.2.5
Signature Revision	2026/01/05-104426
Sigma Revision	r2025-12-01-26-g6fe7343bf
Build Revision	b91539d189f35
Events Lost	0
Events Received	204489
Uptime	~39 minutes
- Log Counts:** A simple table showing log counts.

Category	Count
Info	7
- Modules:** A list of modules categorized as Active or Inactive.
 - Active:** ApplyIDCs, ETWKernelSource, ETWSource, EventlogSource, PollHandles, Rescontrol, Sigma.
 - Inactive:** BeaconHunter, CommandLineMismatchDetector, EtwCanary, LsassDumpDetector, ProcessTamperingDetector, TemporaryDriverLoadDetector.A green button at the bottom of this panel reads "Enable these Modules with the full version of Aurora. Learn More Here!".
- License:** A section showing license details.
 - Owner:** tafeyen337@icloud.com
 - Expires:** 2026/07/09
 - Scanner:** AuroraAgentLite

The Windows taskbar is visible at the bottom, showing the search bar and several application icons. The system tray in the bottom right corner displays the date "26", the time "11:44 PM", and the date "1/16/2026".

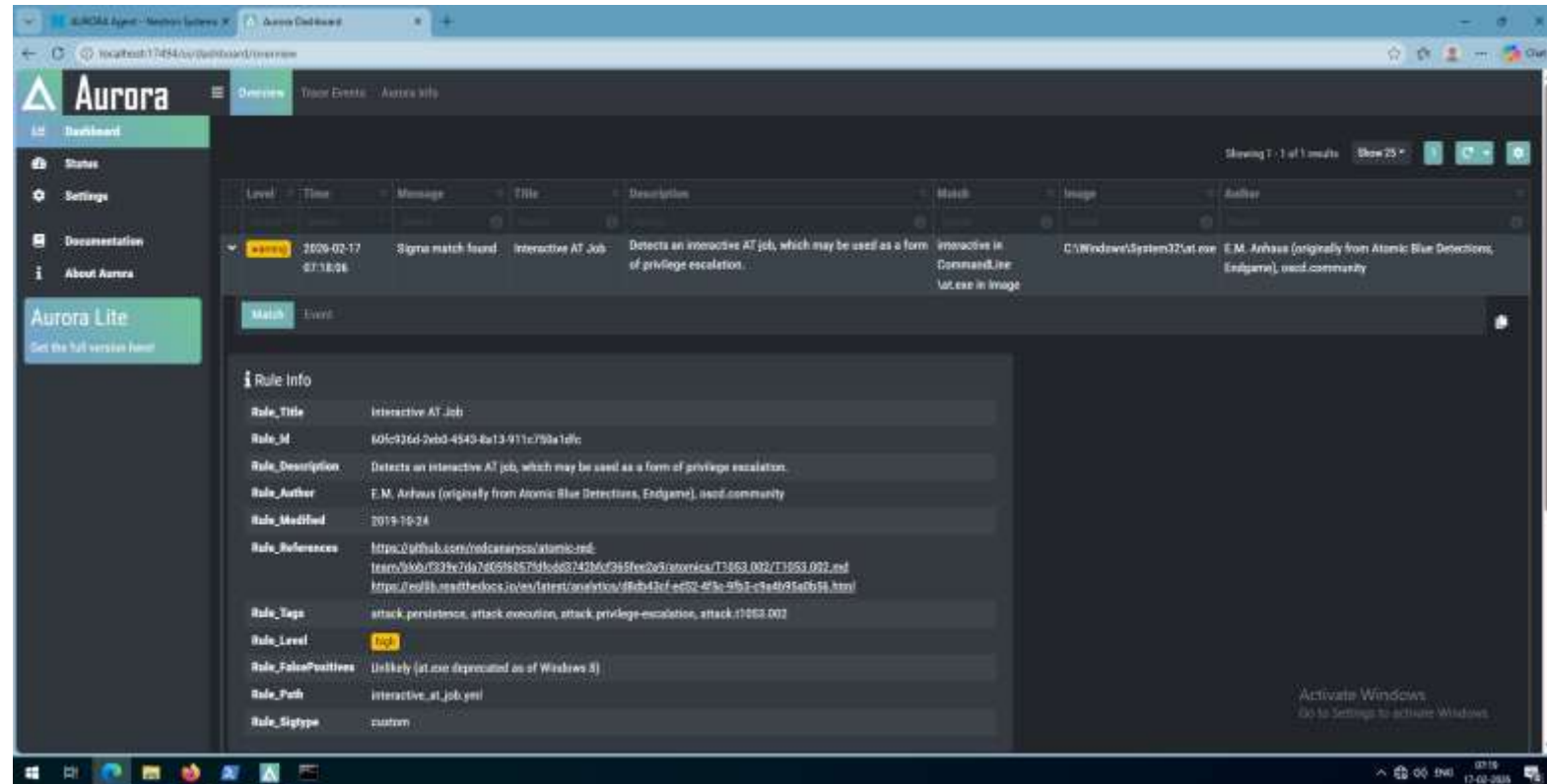
Installation of Aurora Agent

Settings Tab:



Installation of Aurora Agent – Test Installation

1. Just execute command:
 1. at 14:30 /interactive cmd.exe
2. Open Dashboard section and OverView tab
3. You should see one alert with title Interactive AT job



First Sigma Rule

1. We will start with a widely used persistence technique
2. T1547.001 : Boot or Logon AutoStart Execution: Registry Run Keys
 1. Adversaries may achieve persistence by adding a program to a startup folder or referencing it with a Registry run key. Adding an entry to the "run keys" in the Registry or startup folder will cause the program referenced to be executed when a user logs in
3. Execute following commands
 1. Windows + R -> cmd.exe
 2. cd c:\attack
 3. attack01.bat
4. BAT output will also show contents of
HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\CurrentVersion\Run

First Sigma Rule

- Open registry editor by using following commands
 - Windows + R
 - regedit (press enter)
- Open Path
Computer\HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Run
- Check the registry keys – which one seem different ?
- Lets create a basic Sigma rule to capture creation of this registry key

First Sigma Rule

1. Open VSCode / Notepad++ in your Windows 10 VM
2. File -> New File ->
3. File -> Save -> C:\custom_sigma_rules\my_sigma_rule.yml
4. Run following python script to get unique UUID4 ID:
 1. `python c:\sigma_test_repo\get_uuid4.py`
5. Enter Rule values as shown in the next slide

First Sigma Rule

- ID: Use UUID4 generated in previous step (unique for each)
- You can copy from output of attack01.bat the Registry key path
- Keep in mind the same space for same level elements

```
title: My First Sigma Rule Capture New Registry Entry To Autostart
author: xyz
description: "This will detect reg.exe execution with add parameter"
id: 52e815fe-7349-453d-86fe-16fd9f96f5b5
level: high
status: experimental
date: 2026-01-27
logsource:
  category: process_creation
  product: windows
detection:
  selection:
    Image|endswith: '\reg.exe'
    CommandLine|contains|all:
      - 'reg'
      - ' add '
    CommandLine|contains:
      - 'Software\Microsoft\Windows\CurrentVersion\Run'
  condition: selection
```


First Sigma Rule

1. Run following python script to validate our new rule:
 1. `python C:\sigma_test_repo\validate_sigma.py
C:\custom_sigma_rules\my_sigma_rule.yml`
2. Ideal condition you should see output like:
 1. Rule 'C:\custom_sigma_rules\my_sigma_rule.yml' is valid!
3. If you want to check how errors look change the case of letter 'l' in Image|endswith
4. Run the validate_sigma.py script with same command above you will see something like:
 1. `[SigmahqInvalidFieldnameIssue] issue=SigmahqInvalidFieldnameIssue
severity=high description="A field name do not exist"
rules=[52e815fe-7349-453d-86fe-16fd9f96f5b5] field=aaImage
Details: A field name do not exist`

7.1 Rule Testing

1. Newly created rule .yml file is in the C:\custom_sigma_rules directory
2. Now we will stop the Aurora Agent
 1. Press Ctrl+C
 2. Press Y to stop batch job
3. If closed – Re Open Command / PowerShell
4. Restart Aurora agent by executing command:
 1. `.\start_custom_agent.bat`
5. Again Execute attack by using following commands
 1. Windows + R -> `cmd.exe`
 2. `cd c:\attack`
 3. `attack01.bat`

7.1 Rule Testing

- Now open the Aurora Agent management console
- Open the Dashboard section on left menu
- Click on Overview Tab on top to see generated events
- You should see event with same title in rule file

